

Management Information Systems

14. Conclusion

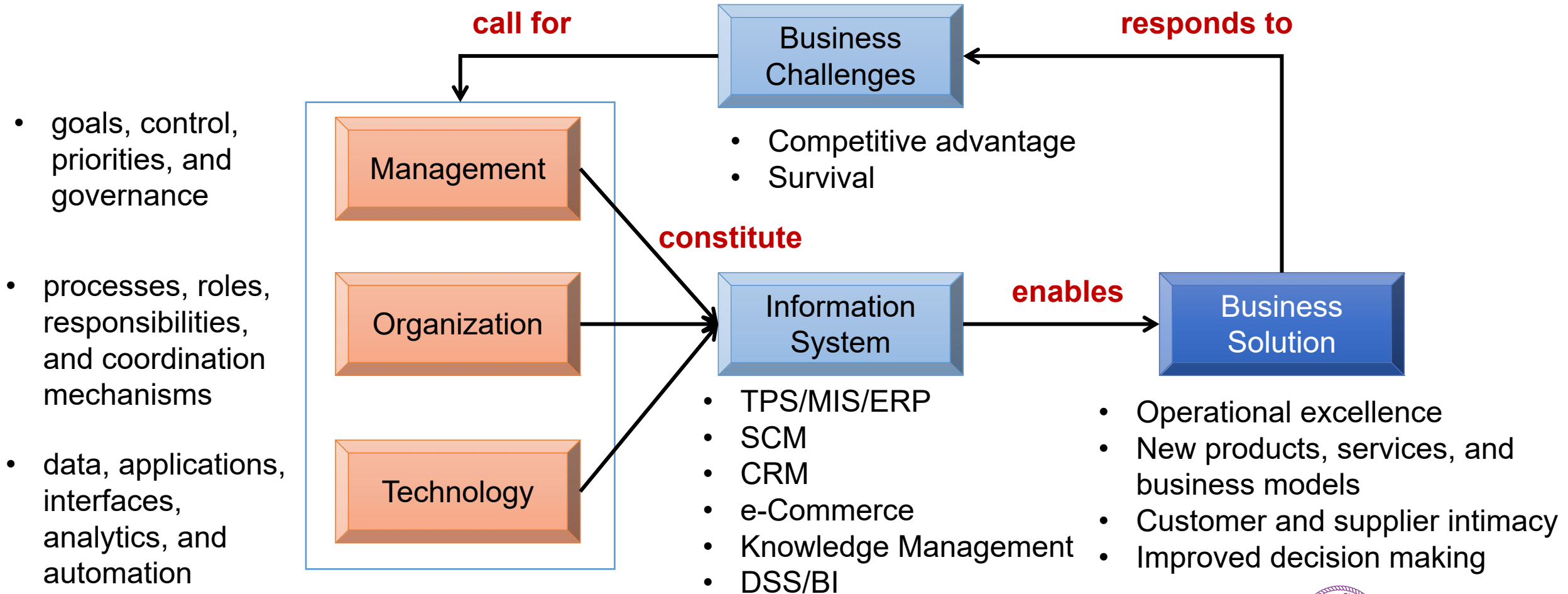
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Overall, IS is NOT just Technology



Course Roadmap: From Principles to Design

- Understanding information systems
 - IS foundations, conceptual dimensions, infrastructure, types of information systems
- Analyzing organizational problems
 - Information gathering, process modeling, object-oriented analysis
- Designing future systems
 - Output, input, databases, HCI, quality assurance, implementation

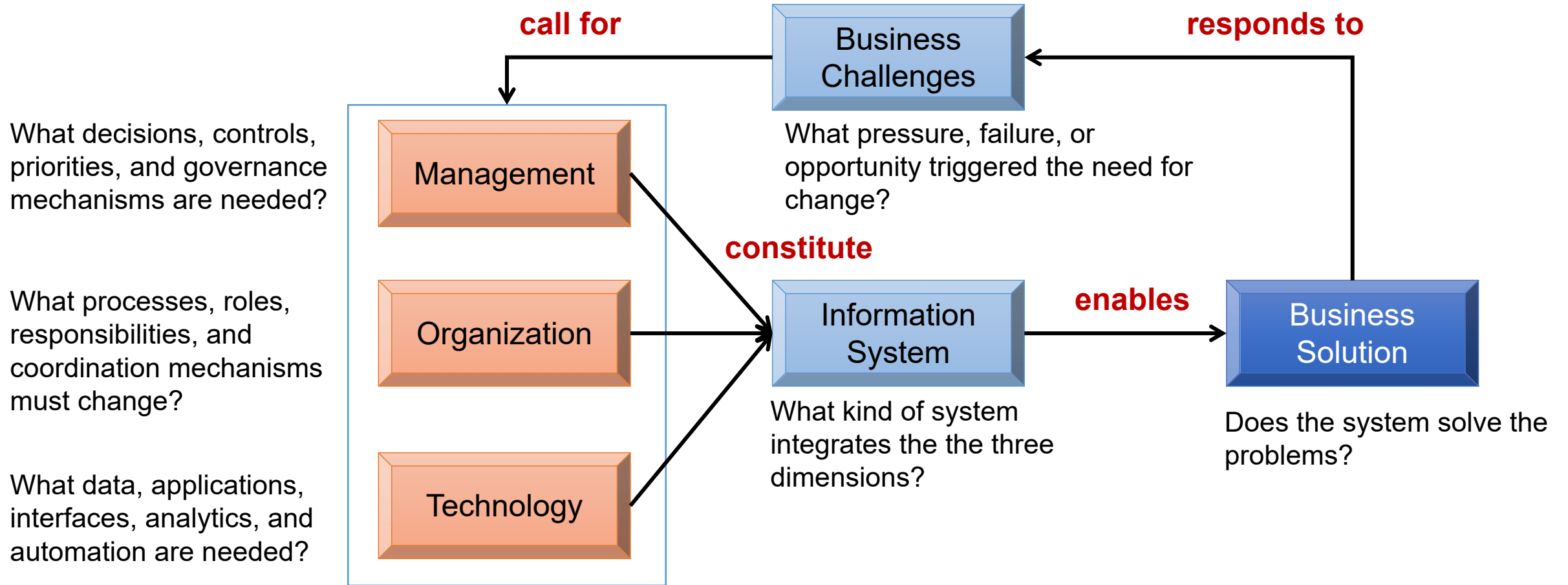


When “We Need a System”

- When organizations say, “We need a new system,” they may actually mean:
 - Information is fragmented.
 - Processes are unclear.
 - Departments cannot coordinate.
 - Data are delayed, duplicated, or inconsistent.
 - People rely on informal workarounds.
 - Managers cannot see operational reality.
 - Customers, employees, or partners experience service failure.
- The essential value of an analyst is not to deliver a system, but to provide a solution
 - A system request is often a symptom.
 - The analyst’s task is to diagnose the deeper organizational problem.



Common Analytical Structure



The MIS Analyst's Mindset

- Information systems are socio-technical systems.
- Technology alone does not solve organizational problems.
- Good analysis begins with understanding current work.
- Requirements emerge through investigation and negotiation.
- Models help clarify complexity.
- Design choices shape behavior, memory, and decision-making.
- Implementation is organizational change.



Trends and Open Issues (I): From Systems to Platforms

- Traditional MIS focused on systems inside organizations.
- Increasingly, organizations operate through:
 - Digital platforms
 - APIs
 - Cloud services
 - Mobile applications
- Implications:
 - Organizational boundaries become more fluid.
 - Coordination extends beyond the firm.
 - Platform governance becomes central.
 - Value creation depends on external participants.
- Core message:
 - The future of MIS is not only internal system design, but ecosystem coordination.



Trends and Open Issues (II): From Data Management to Algorithmic Decision-Making

- MIS is moving from:
 - Recording transactions
- To:
 - Analyzing patterns
 - Predicting outcomes
 - Recommending actions
 - Automating decisions
- Open issues:
 - Data quality, bias, explainability, accountability, overreliance on models, human oversight, and responsible governance.
- Core message:
 - When systems make or shape decisions, MIS becomes increasingly a governance problem, which demands value-driven mechanism design.



Trends and Open Issues (III): Human-AI Relationship

- Important questions:
 - What should humans do?
 - What should AI do?
 - When should AI ask for confirmation?
 - When should humans intervene?
 - How should AI uncertainty be communicated?
 - How should AI actions be audited?
- Human-AI Relationship:
 - Automation (自动化)?: AI performs tasks instead of humans.
 - Augmentation (增强)?: AI helps humans perform tasks better.
 - Guardianship (监护): Humans raise/cultivate (养育) AI agents and the accountability (担责).
- Core message:
 - The central design challenge is no longer only interface design, but responsibility design.



Looking Ahead

- Information systems do not simply support organizations.
- They help constitute organizations.
- They define what can be seen, what can be measured, what can be coordinated, and what can be decided.
- Therefore, to analyze and design information systems is to participate in the design of organizational intelligence.
- We should:
 - Be critical users of systems.
 - Be thoughtful analysts of organizations.
 - Be responsible designers of digital futures.



Exam

- Q&A hours before the final exam
 - 9:00-11:00am, Wednesday, June 17
- Other time
 - Contact me via email, WeChat, or the discussion board whenever you need help
- Exam
 - 9:00-11:00am, Thursday, June 18
 - Location: 6A201



About the Final Exam

- Open-book, open-computer
- **NO network search**
- **NO mobile phones**
- **NO AI tools**
- **NO discussion**
- Format
 - Choice questions (like the review questions): 50-60%
 - Short answer questions and case studies (like the assignments): 40-50%



**That's all,
Thank you ~~~**

